

Saphalya Das

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Education

University of Engineering and Management B. Tech – CSE (AI & ML), Kolkata CGPA: 7.9	2021–2025	Nava Nalanda Santiniketan Secondary & Higher Secondary (10th), India Grade: A	2011–2019
Institute of Engineering and Management M. Tech – CSE, Kolkata CGPA: 7.5 (ongoing)	2025–2027	Nava Nalanda Santiniketan Secondary & Higher Secondary (11th & 12th), India Grade: A	2019–2021

Experience

Research Intern – Backend & AI Systems

IEM Research Foundation, Kolkata, India (On-site) | Nov 2025 – Feb2026

Tech Stack: Python, FastAPI, PostgreSQL, Machine Learning, NLP, Transformers (BERT), REST APIs

Role:

- Led end-to-end backend architecture and development of PsyPredict, a multimodal AI prescreening platform for early mental health assessment using text, behavioral, and emotional data.
- Designed and optimized AI/ML pipelines, fine-tuning Transformer-based models (BERT) for natural language understanding, symptom analysis, and predictive insights.
- Developed scalable Python APIs, real-time inference systems, and data processing workflows for production-ready deployment.
- Collaborated with researchers and psychologists to translate clinical requirements into robust digital solutions while ensuring system reliability and usability.

AI/ML Developer (Contract)

TECHGEN CYBER SOLUTION PVT. LTD. | Mar 2026 – Jun 2026

Tech Stack: Python, FastAPI, PostgreSQL, React, NLP, RAG, FAISS, Knowledge Graphs, LLMs

Role:

- Developed backend services for psychological assessments, automated report generation, patient management, psychologist verification, and role-based access control.
- Built Retrieval-Augmented Generation (RAG) pipelines using FAISS, semantic embeddings, and knowledge retrieval frameworks for assessment interpretation and report enrichment.
- Implemented NLP and LLM-powered modules for narrative analysis, theme extraction, psychological profiling, and automated report generation.
- Designed scalable APIs, database schemas, audit logging mechanisms, transaction workflows, and remote assessment systems.
- Integrated knowledge graph and semantic processing frameworks to enable contextual psychological analysis and decision support.
- Collaborated with psychologists and researchers to convert assessment methodologies into production-ready digital solutions.

Skills

Programming: Python, C, OOP, DSA | **Data:** Data Science, Data Engineering, Data Analysis, Information Retrieval

AI/ML: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, LLMs, RAG, Semantic Search

AI Systems: Knowledge Graphs, Vector Databases, Embeddings, Prompt Engineering, Feature Engineering

Frameworks: FastAPI, React, PostgreSQL, SQLAlchemy, REST APIs, JWT Authentication, RBAC

Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, spaCy, Transformers, Sentence Transformers, FAISS

Development: Backend Engineering, Full-Stack Development, API Development, Database Design, System Design

Architecture: Pipeline Development, Workflow Automation, Assessment Engines, Report Generation Systems

Design: UI/UX Design, Figma, Adobe XD, Product Planning, Prototyping

Projects

GenX – AI Genomic Hybridization Analysis | Computational Biology, BioPython, NumPy, Pandas, ML

Developed a genomic hybridization simulation framework using DNA sequence merging algorithms and predictive biological modelling techniques.

LID-R – AI-Based Object Reshaper | AI, Machine Learning, Mathematical Modeling

Designed an AI-driven system for adaptive reshaping and transformation of geometric and mathematical 3D structures using machine learning techniques.

REGICAM – ML Attendance System | Python, CNN, Computer Vision, NumPy, Pandas

Developed a CCTV-based attendance system using CNN-powered facial recognition for real-time identity detection and automated

attendance tracking.

AI Campus Security System | AI, Computer Vision, Machine Learning

Built an intelligent surveillance platform for anomaly, intrusion, and suspicious activity detection from CCTV feeds using machine learning techniques.

NeurAI(till software prototype) | AI Neural Decoding & Communication System | Python, PyTorch, React, TypeScript, MQTT, Machine Learning

Engineered an end-to-end brain-computer interface (BCI) platform using Raspberry Pi and BioAmp sensors to capture real-time EEG/EMG signals, leveraging PyTorch-based deep learning models for neural signal decoding and real-time text generation through React and MQTT-based communication systems.

Publications

- Multi-Modal AI for Mental Health Prediction and Intervention - International Conference on Web 6.0 and Industry 6.0 (2025)
- LID-R: AI-Based Object Reshaper - Prantik Gabeshana Patrika (ISSN 2583-6706) (2025)
- GenX: AI-Powered Genomic Hybridization Analysis and Prediction - IEEE ICAECT 2026

Awards

- 1st Runner-Up - HACK-A-BIT Pan-India Hackathon (BIT Mesra)
- Chancellor's Award for Technical Contribution to the University - CSE (AI & ML), 2025

Certifications & Workshops

Certifications

- NPTEL - Artificial Intelligence in Industrial Fields
- NPTEL - Developing Soft Skills
- Goldman Sachs - Operations Job Simulation
- Programming Foundations - Coursera
- Introduction to Generative AI - Coursera
- Machine Learning with Python - Coursera
- Cyber Security Fundamentals - University of London
- Gen AI Agents: Transform Your Organization - Google Cloud (Coursera) 2026
- Gen AI: Navigate the Landscape - Google Cloud (Coursera) 2026
- Transformer Models and BERT Model - Google Cloud (Coursera) 2026

Seminars & Workshops

- IoT Security FDP - IIT Guwahati & IEM Kolkata
- Next-Generation Cyber Defense Bootcamp - IEM Kolkata
- ROBO-RACE & ROBO-SOCCER Workshop
- Cloud Computing Workshop - CSI UEM
- International Conference on Web 6.0 and Industry 6.0
- IEEE ICAECT 2026
- Research Promotion Workshop - ACMU, ISI Kolkata
- IIT KGP Research Park - Demonstration of Self-Developed Psychological pre-screening tool (Team Leader, IEM Research Lab)